Notes for Competitive Markets

Example of an Economy

The simplest economy is a day in the life of Robinson Crusoe (character from a book by Daniel Defoe in the 18th century who was stranded on a deserted island) economy, an example that we will revisit.

- $I = J = 1$ and $L = 2$ (coconuts and leisure).
- $X_1 = \mathbb{R}_+ \times [0, 24]$ – Crusoe can consume at most 24 hours of leisure.
- Endowment: $\boldsymbol{\omega} = (0, 24)'$ – no coconuts and 24 hours of leisure.
- Production set: firm (Crusoe himself) can use good 2 to produce good 1:

$$Y_1 = \{(x_1, x_2) : x_1 \leq f(24 - x_2)\}$$

  where f is strictly increasing and concave. Can think of $24 - x_2$ as z, the firm input.

Feasibility

An allocation $(\boldsymbol{x}_1, \boldsymbol{y}_1)$ is feasible if:

$$x_{11} = 0 + \underbrace{f(24 - x_{21})}_{\text{Firm output}}$$

$$x_{21} = 24 - \underbrace{(24 - x_{21})}_{\text{Firm input}}$$

Example of a Private Ownership Economy

- Crusoe owns all of the endowment: $\omega_{11} = 0$ and $\omega_{21} = 24$.
- Crusoe owns the firm, $\theta_{11} = 1$.
- The price vector $\boldsymbol{p}$ is the price of coconuts, p_1 and the price of leisure, p_2 (wage rate).
- The budget set is then all x_{11} and x_{21} satisfying

$$p_1 x_{11} + p_2 x_{21} \leq 24 p_2 + p_1 f(24 - x_{21}) - p_2 (24 - x_{21})$$

  RHS: Crusoe sells all his endowment and gets the revenue from selling coconuts, but has to pay himself a wage of p_2 for hours worked. LHS: Crusoe has to pay for the coconuts and leisure he consumes.

- Relating consumer problem to how univariate optimization problem with single inequality constraint looks in book's appendix:

$$\max_{x} f(x) \text{ subject to } h(x) \leq c$$

KT conditions: $f'(x) = \lambda h'(x)$ and $\lambda(h(x) - c) = 0$. Here, $h(x) = -x$ and $c = 0$:

$$\max_{x_i} \ \phi_i(x_i) - p^\star x_i \text{ subject to } x_i \geq 0$$

KT conditions: (i) $\phi_i'(x_i) - p^\star = -\lambda$ and $\lambda x_i = 0$. So $\phi_i'(x_i) - p^\star + \lambda = 0$, where $\lambda = 0$ if $x_i > 0$ and $\lambda > 0$ if $x_i = 0$ (so $\phi_i'(x_i) \leq p^\star$).

- Why do we choose m_i from $\mathbb{R}$ and not $\mathbb{R}_+$? This avoids the boundary conditions. A consumer will choose x_i to satisfy $\phi_i'(x_i) \leq p$, with equality if $x_i > 0$. However, if the consumer's endowment is not enough to afford the x_i satisfying $\phi_i'(x_i) = p$, then we need to always include cases for that. This makes this a bit simpler.

- The equilbrium conditions are $I + J + 1$ equations which allow us to solve for the $I + J + 1$ unknowns (quantities and price).

- Why do we need the $c_j'(q_j) \to \infty$ as $q_j \to \infty$? To make sure supply and demand cross (avoid asymptote sitation).

SWT

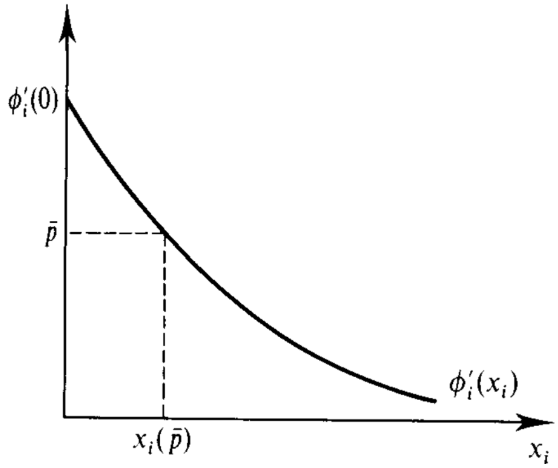
- We can redistribute endowments of the numeraire without affecting the production and demand for good ℓ.

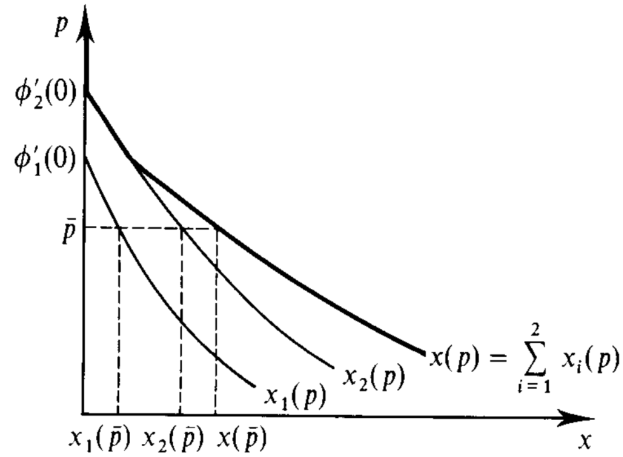
- Therefore we can achieve any point on the utility possibility frontier as part of a competitive equilibrium by redistributing an appropriate amount of the numeraire to consumers.

- For any Pareto optimal levels of utility $(u_1^\star, \ldots, u_I^\star)$, there are transfers of the numeraire commodity $(T_1, \ldots, T_I)$ satisfying $\sum_i T_i = 0$, such that a competitive equilibrium reached from the endowments $(\omega_{m1} + T_1, \ldots, \omega_{mI} + T_I)$ yields precisely the utilities $(u_1^\star, \ldots, u_I^\star)$.

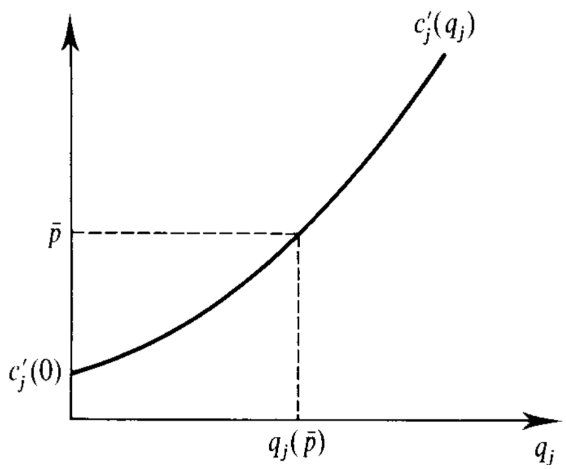
Partial Equilibrium

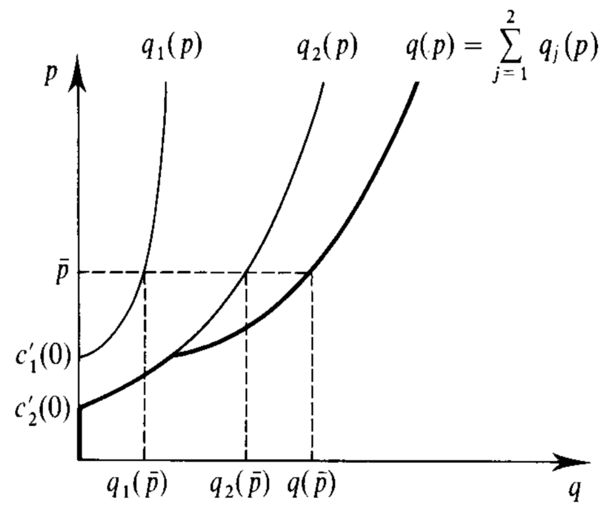
Demand and aggregate demand:



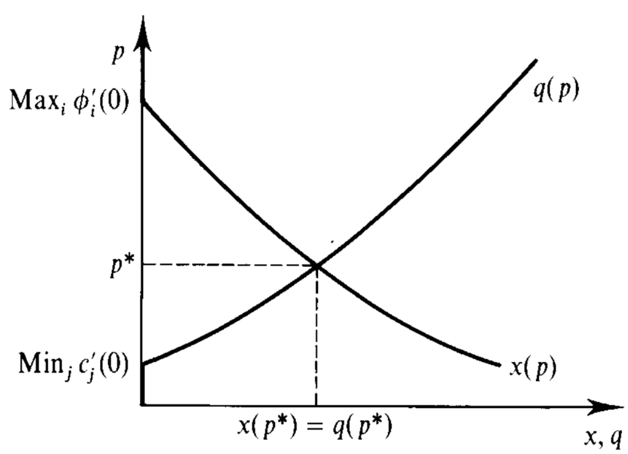


Supply and aggregate supply:

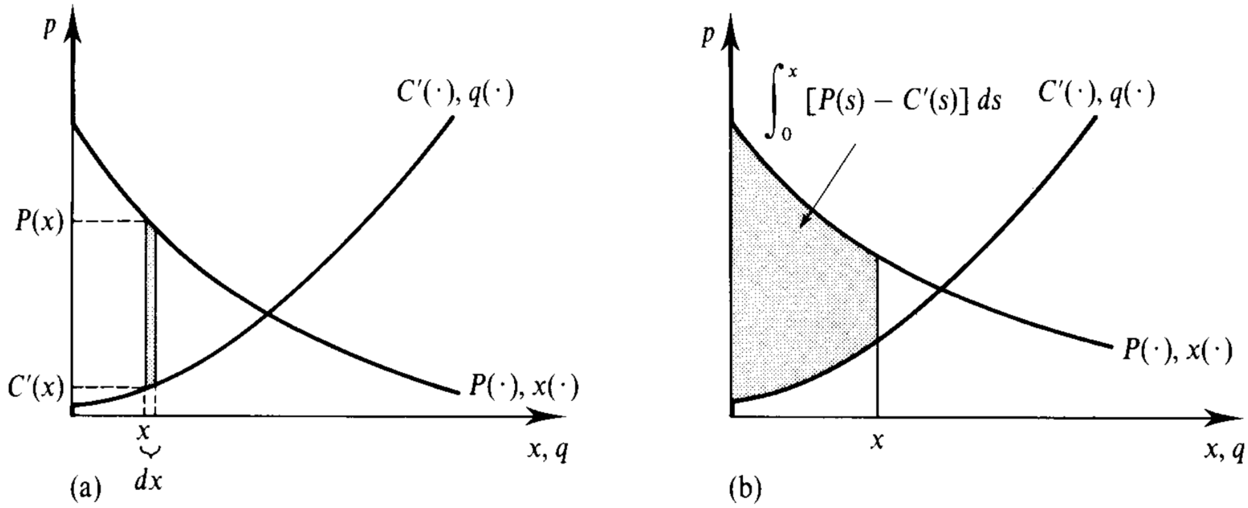




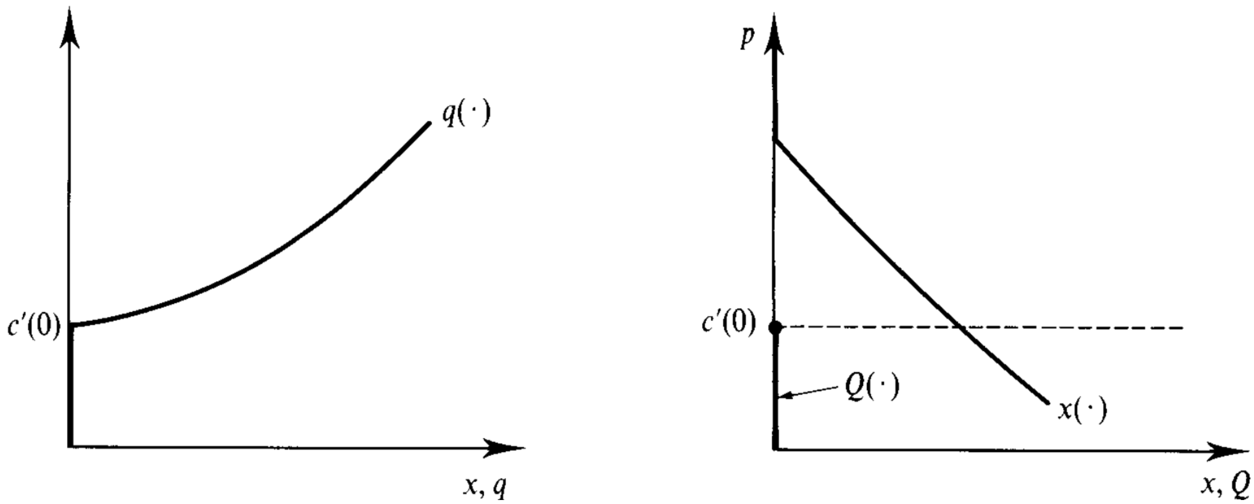
Equilbrium:



Marshallian Aggregate Surplus:



Strictly convex costs:



Positive efficient scale: